

ENGINEERING METROLOGY & INSTRUMENTATION LAB

IV B. Tech. I Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6ME44	PCC	0	0	2	1	40	60	100

COURSE OVERVIEW:

This course is introduced to create awareness on various mechanical measuring instruments. This course intended to learn different types of transducers.

COURSE OUTCOMES:

At the end of the course, the student should be able to:

1. Perform the measurements Length, Depth, Diameter measuring using Vernier calipers & micrometer.
2. Perform Bore measurement using bore gauge.
3. Use the gear teeth caliper for checking the chordal addendum and chordal height of spur gear.
4. Ability to calibrate and analyze the measuring system for the measurement of displacement and pressure.
5. Ability to calibrate and analyze the measuring system for the measurement of speed.

LIST OF EXPERIMENTS:

1. Measurement of outside & inside diameter using vernier callipers.
2. Measurement of outside & inside diameter using micrometer.
3. Height measurement by Vernier height gauge.
4. Angle measurement by using Bevel protractor.
5. Angle measurement by using Sine bar & Slip Gauges.
6. Calibration of Pressure Gauges.
7. Study and calibration of LVDT transducer for displacement measurement.
8. Calibration of capacitive transducer for angular displacement.
9. Study and calibration of photo and magnetic speed pickups for the measurement of speed.
10. Study and calibration of a rotometer for flow measurement.